Principal Component Analysis (PCA) Report: Yield Curves and Forward Rates

Financial Risk Analysis

January 17 2025

1 Overview

This paper analyzes yield curves for Canadian Government bonds during January 2025. It uses linear interpolation to construct the yield curve. One-year forward rates are obtained from interpolated bootstrapped spot curves. Finally, the principal component analysis presents the impact of the first component (parallel shift).

The yield curves are U-shaped, and the principal component analysis of the yield curve indicates a parallel upward shift. This means that whatever medium-term easing and long-term tightening hold, the entire rate environment is still up.

The forward one-year rate curves slope up, suggesting an increase in one-year rates. But the PCA first component suggests a downward shift. This combination means that the market has revised the entire rate environment lower, while still expecting interest rates to rise gradually over time.

2 Data Collection

Using businessinsider.com website I selected ten bonds with maturities on September 1st and March 1st with first bond maturing on March 1st 2025 and the last bond maturing on September 1st 2029 which are: CAN 1.25 Mar 25, CAN 0.5 Sept 25, CAN 0.25 Mar 26, CAN 1.00 Sept 26, CAN 1.25 Mar 27, CA 2.75 Sept 27, CAN 3.50 Mar 28, CAN 3.25 Sept 28, CAN 4.00 Mar 29, CAN. I selected these bonds for numerous reasons. First, a benefit was that each bond was the sole bond that matured on the specified date, thereby avoiding the need to choose between 2 bonds with different prices maturing on the same date. Secondly, all maturities were organized chronologically, with a six-month gap between subsequent maturities. These factors allow for bootstrapping to be performed without interpolations, creating more reliable results.

3 Pseudocode

```
1) LOAD + PREP DATA
- Read "Canadian Bond Data.csv" into DataFrame
- Convert "Issue date" and "Maturity Date" to datetime
- Convert Coupon from "x%" string to decimal
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- Filter bonds to maturities on Mar 1 or Sep 1; sort by maturity; keep maturities  2029
- Identify close price columns: close_cols = columns starting with "Close"

2) HELPERS + CLASSES
Helper: subtract_6_months(date) -> date shifted back 6 months

Class Bond: store ISIN, IssueDate, Coupon, Maturity, Face

Class Bond_Analytics(Bond): add CurrentDate, ClosePrice
- payments_due(): build semiannual cashflows from maturity backward by 6M until
    ↪ CurrentDate
- last_Coupon_Date(): 6M before first future cashflow (or IssueDate if later)
- Dirty_Price(): ClosePrice + accrued interest
- PV(y): discount cashflows using semiannual compounding
- Yield(): solve PV(y) = DirtyPrice via Newton root find

Class Yield_Curve
- Insert_Bond(b): store b.Yield() with b.MaturityDate
- Linear_interpolation(d, maxYears): interpolate yields to grid 0..maxYears step 0.5

Class Spot_curve
- Bootstrap_Insert(b):
    if first bond: spot  b.Yield()
    else: discount earlier coupons with existing spots and solve new spot from remaining
        ↪ PV
- Linear_interpolation(d, maxYears): interpolate spots to grid 0..maxYears step 0.5

Class y1_Forward_Curve
- Build_Forward_Curve(tenors, spots): compute forward rates between adjacent tenor points

3) YIELD CURVES + PCA
- For each Close_Date in close_cols:
    d = parse date from column name
    build Yield_Curve by computing each bonds Yield at date d
    interpolate yields on 0..5y (step 0.5), plot, store in YTM_matrix
- PCA on yields:
    X = array(YTM_matrix)
    log_returns = diff(log(X)) over time
    cov = covariance(log_returns)
    eigen-decompose, sort, print top 5 PCs

4) SPOT CURVES + FORWARDS + PCA
- For each Close_Date:
    bootstrap Spot_curve by maturity order using bond dirty prices
    interpolate spots on 0..5y (step 0.5), plot, store in Interpolated_Spot_Matrix
- For each interpolated spot curve:
    build forward curve from adjacent interpolated spots
    plot selected points (program slices to approximate 1Y forwards)
    store selected forward rates in Forward_matrix
- PCA on forwards:
    X = array(Forward_matrix)
    log_returns = diff(log(X)) over time
    cov = covariance(log_returns)
    eigen-decompose, sort, print top 4 PCs
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```
END
```

4 Yield Curve Analysis

The linearly interpolated yield curve is U-shaped. This means that short-term interest rates are expected to decline and long-term rates to increase. The decline in short-term rates reflects an expectation of monetary loosening. Higher long-term rates indicate increased uncertainty and a term premium.

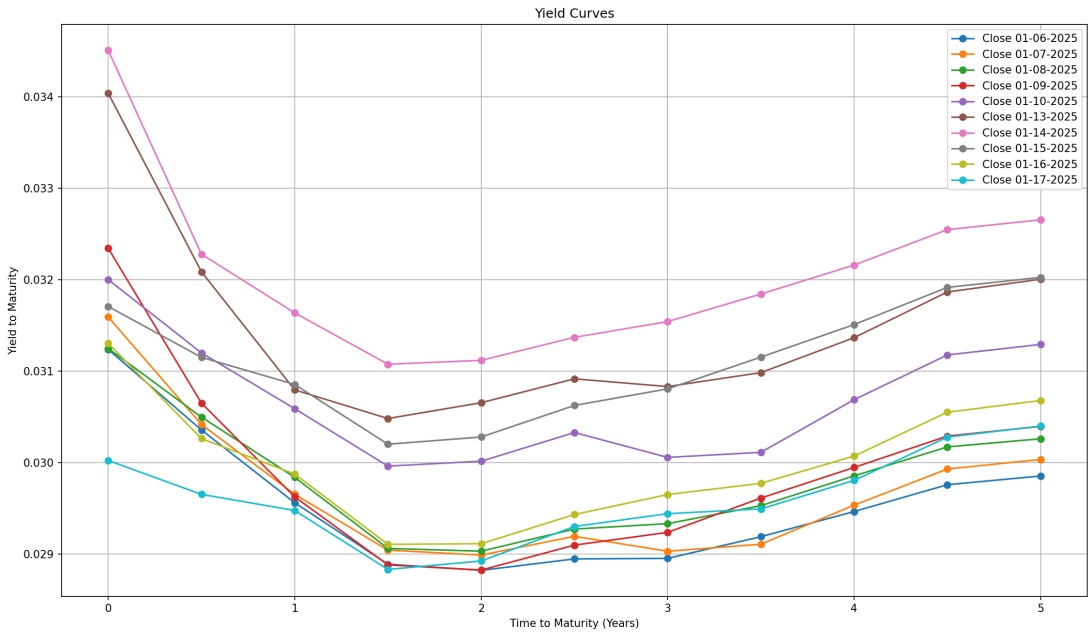

Figure 1: Yield Curves

The **principal component analysis**. The first factor explains **80.5%** of the variance as indicated by the first eigenvalue. It means that the parallel shift upwards is the most impactful factor. The upward movement is confirmed by the positive eigenvector components.

Table 1: Yield Curve PCA Summary

Factor	**Eigenvalue**	**Expl. Var**	**Interpretation**
PC1	5.748×10^{-3}	80.49%	Parallel Shift (Level)
PC2	1.189×10^{-3}	16.65%	Twist (Slope)
PC3	1.352×10^{-4}	1.89%	Butterfly (Curvature)
PC4	4.431×10^{-5}	0.62%	Residual
PC5	1.505×10^{-5}	0.21%	Noise

Table 2: Yield Curve Eigenvectors (Factor Loadings)

Factor	Eigenvector										
PC1	0.390	0.271	0.262	0.295	0.304	0.295	0.292	0.302	0.301	0.294	0.292
PC2	0.885	0.128	-0.177	-0.151	-0.176	-0.204	-0.115	-0.075	-0.139	-0.136	-0.126
PC3	0.069	0.090	0.168	0.329	0.380	0.376	-0.115	-0.607	-0.366	-0.168	-0.145
PC4	0.092	-0.303	0.839	0.141	-0.215	-0.268	0.027	0.045	0.012	-0.138	-0.191
PC5	0.058	-0.121	0.115	-0.039	-0.223	0.006	-0.531	-0.411	0.236	0.444	0.462

5 Forward Rate Curve Analysis

We construct a spot curve using bootstrapping. Then we derive the forward rate from the linearly interpolated phase bootstrapped spot curve rates. The forward rates are upward slopping indicting that after the easing phase, the one-year rates are expected to rise again.

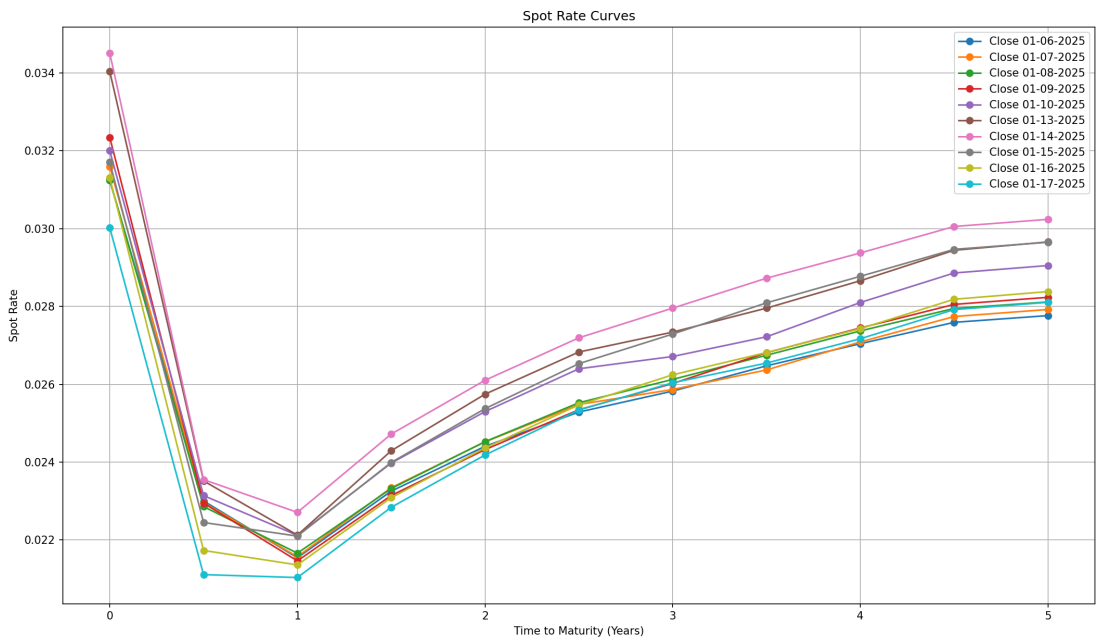

Figure 2: Spot Rate Curve

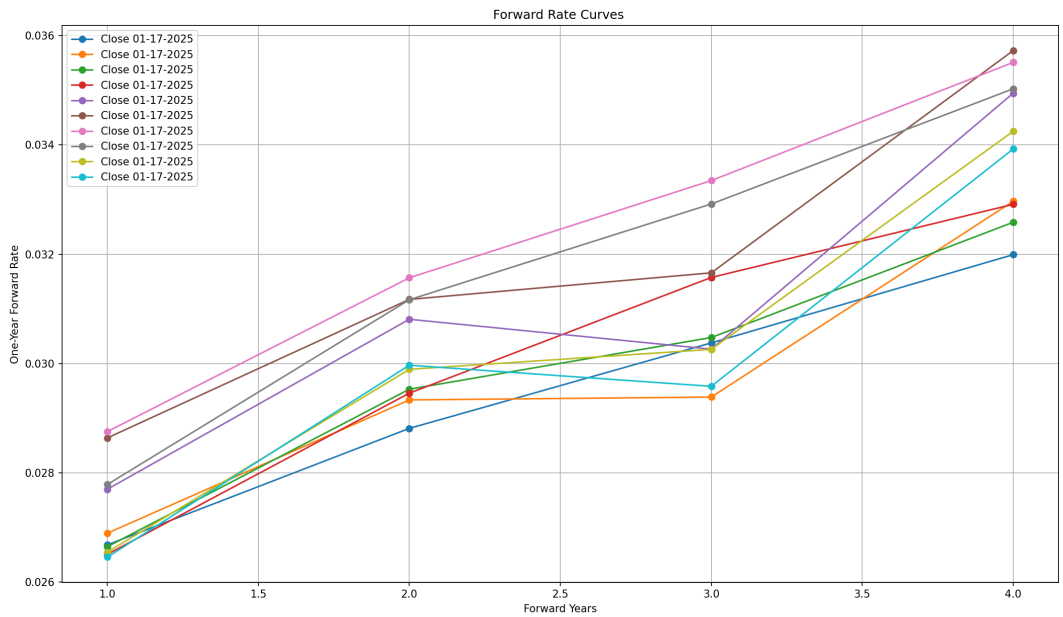

Figure 3: One Year Forward Rate Curves

The **principal component analysis**. The first factor explains **58.94%** of the variance as indicated by the first eigenvalue. It means that the parallel shift downwards is the most impactful factor. The downwards movement is confirmed by the negative eigenvector components.

Factor Loadings

Table 3: Forward Rate Factor Loadings

Factor	**Eigenvalue**	**Variance**	Eigenvector
PC1	0.00254	58.94%	[-0.391 -0.309 -0.844 -0.199]
PC2	0.00165	38.29%	[-0.487 -0.398 0.510 -0.588]
PC3	0.00007	1.61%	[0.075 -0.800 0.120 0.583]
PC4	0.00005	1.16%	[-0.777 0.327 0.117 0.524]